

Recipes_For_AllAboutFood – Alexa Skill Documentation

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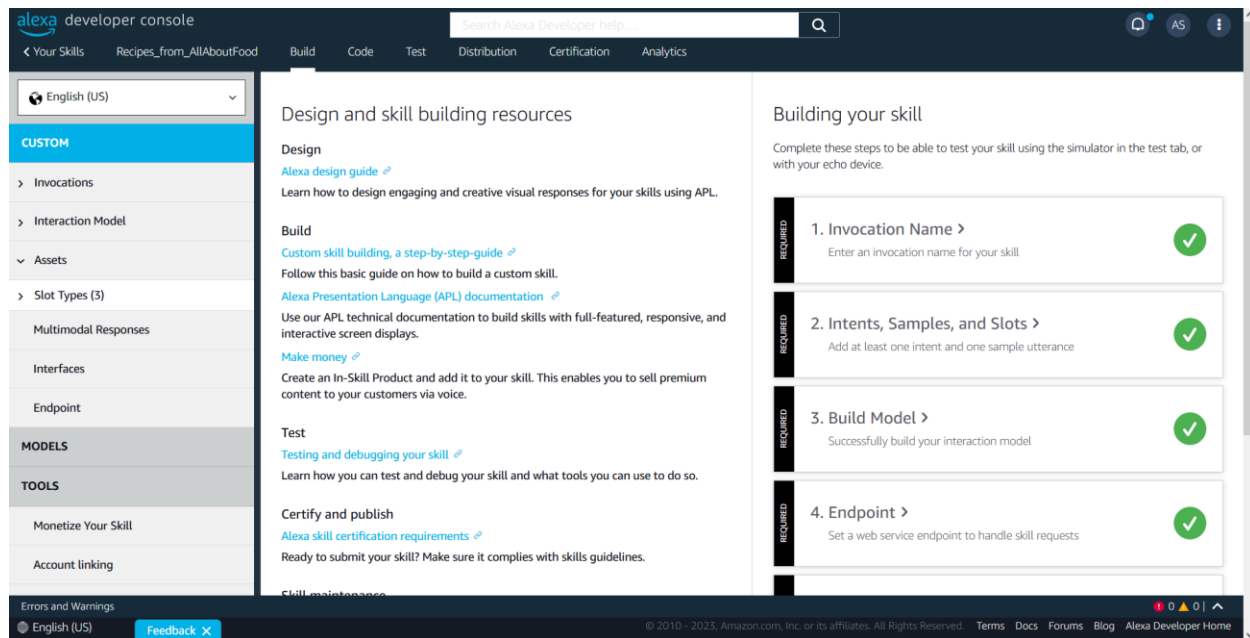
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Introduction:

This guide will be going through each section of the Alexa Skill console. The first sections will each cover the main tabs at the top of the screen: Build, Code, Test, Distribution, Certification, and Analytics.

Build:

This should be where the Alexa Skill first opens up to.



To go over each section along the right side of the screen.

Invocations:

This section is where the skill invocation name is set, which is what is used to open the skill. Additionally, after the skill has been published, you can also add additional skill launch phrases, which can be used to start the skill without stating the skill name.

Interaction Model:

The main focus of this section is the Intents, which are the various commands that the skill recognizes after it has been opened. For example, the OpenPublicRecipeIntent defines what users can say in order to open a public recipe. This is what you would want to edit in order to add more ways to invoke an action or if you wanted to add a brand new command.

It is important that if you make any changes to the intents, you must rebuild the model. The build model button should appear just below the various tabs when you have the Interaction Model section open.

Assets:

The main focus under the Assets section is the Multimodal Responses. These are the various screens that are shown on the Alexa Display while using the skill. Under actions each of these screens can be edited using APL. Creating and Editing screens can be fairly complicated, amazon provides a number of premade screens that can generally be adjusted to a specific purpose. A more detailed description of editing APL can be found in the amazon documentation:

<https://developer.amazon.com/en-US/docs/alexa/alexa-presentation-language/create-and-edit-apl-document.html>.

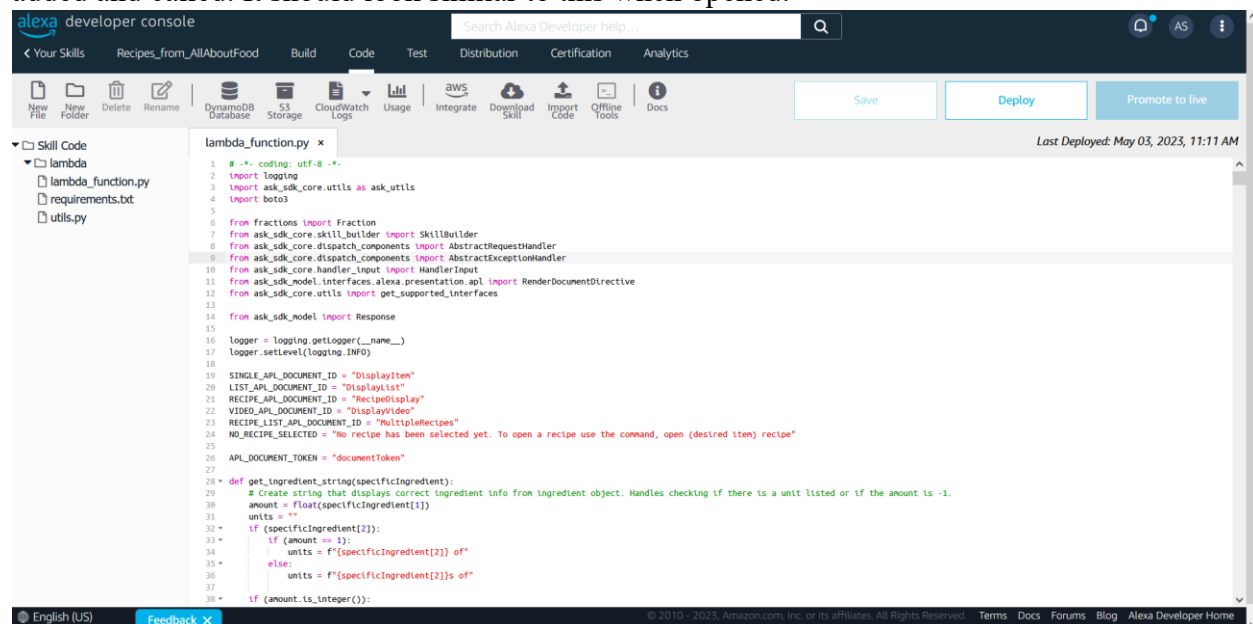
Also under actions, you can see example code that will show you how to implement the screen into your skill, with provided python code.

Account Linking:

The other important part of the Build tab is the account linking settings. None of this should ever need to be changed, but the Web Authorization URI is the address of the firebase hosted authorization site which links the skill to the app accounts. The client ID is used by the site to ensure that the skill is attempting to sign in and to reject random access to the address.

Code:

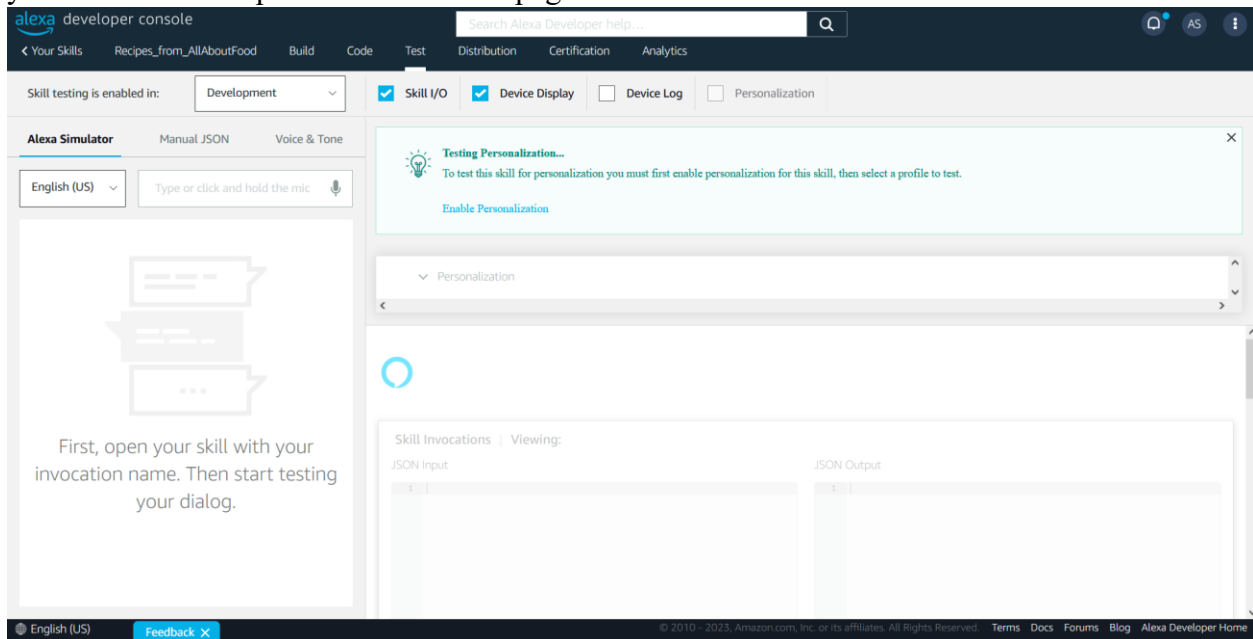
This section should open up to the `lambda_function.py`, which is where all of the main code is added and called. It should look similar to this when opened.

The screenshot shows the Alexa Developer Console interface. At the top, there's a navigation bar with tabs for 'Your Skills', 'Recipes_from_AllAboutFood', 'Build', 'Code' (which is active), 'Test', 'Distribution', 'Certification', and 'Analytics'. Below the navigation bar is a toolbar with icons for file operations (New File, New Folder, Delete, Rename), AWS services (DynamoDB, S3, CloudWatch, Usage), and other tools (Integrate, Download Skill, Import Code, Offline Tools, Docs). On the left, a sidebar shows the project structure: 'Skill Code' > 'lambda' > 'lambda_function.py' (selected), 'requirements.txt', and 'utils.py'. The main area displays the code for 'lambda_function.py'. The code is a Python script that imports various modules like logging, boto3, and the Alexa SDK. It defines a skill builder and a request handler. The code includes comments and logic for handling different intents like 'DisplayItem', 'DisplayList', 'RecipeDisplay', 'RecipeVideo', and 'MultipleRecipes'. It also includes a function 'get_ingredient_string' that formats ingredient information. The code is well-commented and structured. At the bottom of the console, there's a status bar showing 'English (US)', a 'Feedback' button, and copyright information: '© 2010 - 2023 Amazon.com, Inc. or its affiliates. All Rights Reserved. Terms Docs Forums Blog Alexa Developer Home'.

All of this code should be fairly well explained with function names and comments. If you are adding a new Intent, then the first thing you should do is create a new intent handler, the basic format of which can be found in the existing code. After adding the new handler, you must then go to the bottom of the file and add it to the skill builder in order for it to be accessible by the skill. After you make any changes to the code, you must deploy it before it takes effect.

Test:

This page is where you can test the Alexa skill in-browser, on the left side of the screen is where you can enter example commands. The page should look like this.



The first thing you would usually do is use a command like “open all about food” and then you should have all of the normal skill functionality, such as opening recipes or reading directions. In order to view an example display, you need to make sure it is checked at the top and then you will probably need to scroll down near the bottom of the right display box. This is generally the easiest and quickest way of testing changes to the skill. Before testing with this, you need to make sure that you deployed the code or built the model, depending on what changes you made. If the console is left open for some time it can occasionally behave irregularly, which can be fixed by simply reloading the page.

Distribution:

This section covers most of the skill information that it will be published with, such as the skill name and description, as well as links to the privacy policy and terms of use.

alexa developer console

Search Alexa Developer help

< Your Skills recipes_for_allaboutfood Build Code Test Distribution Certification Analytics

Skill Preview

English (US)

Privacy & Compliance

Availability

English (US) Store Preview

Tell us how your skill should appear to English (US) customers.

Copy from an existing locale

Primary Details

Public Name * ?

recipes_for_allaboutfood

24/50 characters

One Sentence Description * ?

Sample Short Description

24/160 characters

Detailed Description * ?

Sample Full Description

23/4,000 characters

English (US) Feedback

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English (US):

The tab should open to the English Store Preview, where you provide the information for publishing it to US stores. If you added availability to different regions or locals, they would be listed underneath this one. This is where you would enter basic information of the skill and provide links to the Privacy Policy and Terms of Use.

Privacy & Compliance:

This tab is where you mark information about your skill such as whether users can make purchases with it or whether your skill contains advertising.

Availability:

This tab is where you set who can access the skill as well as how you would invite beta testers to have access to the skill before it is published.

Certification:

This section is how you check whether your skill is ready to be published, and if not, what you still need to do. It also will create a version history of your skill that will update when you publish new versions.

The screenshot shows the 'Validation' page in the Alexa Developer Console. The left sidebar contains links for 'Validation', 'Submission', 'Version History', and 'A/B Testing'. The main content area is titled 'Validation' and features a 'Run' button. A red banner indicates '21 Fixes Required'. Below this, a list of 'Required Fixes' is provided, detailing issues such as missing selections for personal information, skill categories, purchase permissions, age targeting, advertising, and export compliance. The page also shows a section for 'English (AU)' with specific fixes for the en-GB locale and privacy policy URL. The footer includes the language selector (English (US)), a feedback button, and copyright information.

alex developer console

Search Alexa Developer help...

< Your Skills recipes_for_allaboutfood Build Code Test Distribution Certification Analytics

Validation

Run

21 Fixes Required

Required Fixes

- A selection is missing for the question 'Does this Alexa skill collect users' personal information?'. In the Distribution tab, go to the Privacy & Compliance section to provide an answer.
- The Category is missing for this skill. In the Distribution tab, go to Skill Preview to add a skill category.
- A selection is missing for the question 'Does this skill allow users to make purchases or spend real money?'. In the Distribution tab, go to the Privacy & Compliance section to provide an answer.
- A selection is missing for the question 'Is this skill directed to or does it target children under the age of 13?'. In the Distribution tab, go to the Privacy & Compliance section to provide an answer.
- A selection is missing for the question 'Does this skill contain advertising?'. In the Distribution tab, go to the Privacy & Compliance section to provide an answer.
- Confirmation for 'Export compliance' is missing. In the Distribution tab, please go to the Privacy & Compliance section to resolve the issue.

English (AU)

- A Large Skill Icon is missing for the en-GB locale. In the Distribution tab, go to Skill Preview for en-GB to add a large skill icon.
- The privacy policy URL is missing for the en-US locale. The privacy policy URL is required for skills that collect user information (e.g. account linking). In the Distribution tab, go to Skill Preview for en-US to add the privacy policy URL.

English (US) Feedback X

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Analytics:

This section covers analytic information about the skill such as the number of sessions, unique customers, and what intents are being used. I will not be covering this in detail, as it is mostly for use after the skill has been published.

The screenshot shows the 'Analytics' page in the Alexa Developer Console. The left sidebar contains links for 'Summary', 'Skill Quality Coach', 'CUSTOM', 'Customers', 'Sessions', 'Utterances', 'Intents', 'Retention', 'Interaction Path', 'Distribution', 'PERFORMANCE', 'Performance Summary', 'Intent Confidence', and 'Endpoint Latency'. The main content area is titled 'Summary' and features a 'Jump to analytics for:' dropdown set to 'recipes_for_allaboutfood'. Below this, there are filters for 'Skill Stage' (Development), 'Time Interval (UTC)' (Today, Yesterday, Last 7 Days, Last 30 Days, Custom Range), and 'Aggregation Period' (Daily). The 'Intent' filter is set to 'All'. The 'Custom Model' section displays metrics for 'Sessions' and 'Unique Customers' for the 'Last 7 Days'. The 'Skill Activation' section displays metrics for 'User Enablements' and 'Account Linking Completion Rate' for the 'Last 7 Days'. The footer includes the language selector (English (US)), a feedback button, and copyright information.

alex developer console

Search Alexa Developer help...

< Your Skills recipes_for_allaboutfood Build Code Test Distribution Certification Analytics

English (U.S.)

Jump to analytics for: recipes_for_allaboutfood

Summary

Skill Quality Coach Beta

Development Today Yesterday Last 7 Days Last 30 Days Custom Range

Aggregation Period Daily

Intent All

Custom Model

Sessions

Total for Last 7 Days ▲ 17 (previous 7 days had 0)

Maximum per Day 17 on 05/12/2023 UTC

Average per Day 2.43

Unique Customers

Total for Last 7 Days ▲ 2 (previous 7 days had 0)

Skill Activation

User Enablements

Total for Last 7 Days ▲ 2 (previous 7 days had 0)

Maximum per Day 1 on 05/11/2023 UTC

Average per Day 0.5

Account Linking Completion Rate

English (US) Feedback X

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Outside Links:

This section will cover how I set up the links from the Alexa Skill to DynamoDB and to the Firebase authorization site.

DynamoDB:

This link is what I used for the general process: <https://developer.amazon.com/en-US/docs/alexa/hosted-skills/alexa-hosted-skills-personal-aws.html>

In summary, from the code page of the Alexa skill, I hit the integrate button near the top of the screen to get the skills ARN.

I then went to the AWS IAM dashboard and selected the roles tab on the left. I then added the role: `AlexaSkillDynamoAccess`, with the skill ARN in its trusted entities. I then copied the ARN of that role and use it in the `LaunchRequestHandler` function of the skill.

Firebase Authorization:

This was set up in the Build section under Account Linking. I did not find any guide for this so I cannot guarantee this was the best way to do it. In the github under `AllAboutFoodAuthorization`, there is the website that is hosted on Firebase. The skill connects to the site and passes the `ClientID` in the URL. The site verifies the ID and then if accepted allows the user to sign in. After they sign in, it then redirects back to amazon with an access token for the user that just signed in.